IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

WAVELETS, REPRESENTATION LEARNING AND THEIR APPLICATIONS

Friday 19 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : A. Manikas

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Special Information for the Invigilators: NONE

Information for Candidates:

Sub-sampling by an integer N

$$x_{\downarrow N}[n] \longleftrightarrow \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega-2\pi k)/N}) = \frac{1}{N}\sum_{k=0}^{N-1} X(W_N^k z^{1/N}),$$

where

$$W_N^k = e^{-j2\pi k/N}.$$

Poisson Summation Formula

$$\sum_{n=-\infty}^{\infty} \varphi(t-nT) = \frac{1}{T}\sum_{k=-\infty}^{\infty} \hat{\varphi}\left(\frac{2\pi k}{T}\right) e^{-j2\pi kt/T}$$

The Questions

1. Consider the two-channel filter bank of Figure 1a.

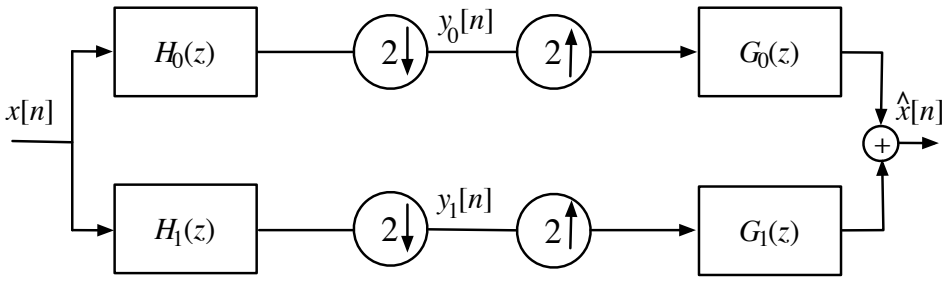


Figure 1a: Two-channel filter bank.

(a) Express $\hat{X}(z)$ as a function of $X(z)$ and the filters. Then derive the two perfect reconstruction (PR) conditions the filters have to satisfy. [5]

(b) Consider

$$P(z) = H_0(z)G_0(z) = \frac{1}{\pi}(z + \pi + z^{-1}).$$

i. Show that $P(z)$ satisfies the half-band condition: $P(z) + P(-z) = 2$. [5]

ii. Factorize $P(z)$ and assign one zero of $P(z)$ to $G_0(z)$. Choose $G_0(z)$ to be minimum phase. [5]

iii. Design the other filters in order to have a perfect reconstruction biorthogonal filter bank. [5]

iv. Now, consider the limit function

$$\hat{\varphi}(\omega) = \lim_{J\to\infty} \prod_{k=1}^{J} M_0\left(\frac{\omega}{2^k}\right),$$

where $M_0(\omega) = \frac{G_0(e^{j\omega})}{\sqrt{2}}$ and $G_0(z)$ is the filter you found in Part-ii. What can you say about convergence of the above limit? [5]

2. (a) Consider the system shown in Figure 2a, where $H(z)$ is an ideal low-pass filter with cut-off angular frequency $\omega = \pi/4$. Sketch and dimension the three spectra $Y_1(e^{j\omega}), Y_2(e^{j\omega})$ and $Y_3(e^{j\omega})$ assuming that $x[n]$ has the spectrum shown in Figure 2b.

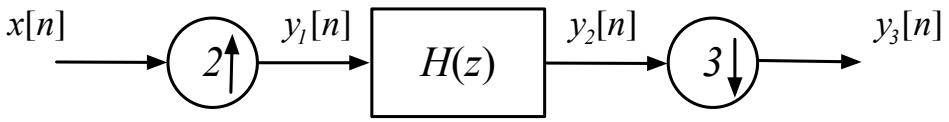


Figure 2a: Multi-rate system with low-pass filter $H(z)$.

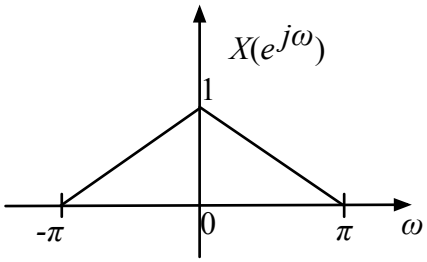


Figure 2b: Discrete-time Fourier transform of $x[n]$.

[5]

(b) Consider the system

$$y = HD_2GD_2HFD_3x$$

where G, H and F are the infinite matrix representations for filtering and D_k is the matrix representation of downsampling by k with $k = 2, 3$.

i. Draw a block diagram of the system. [5]

ii. Derive an equivalent system consisting of a single filter block and a single downsampling matrix. Write the z-transform of the equivalent system as a function of $G(z)$, $H(z)$ and $F(z)$. [5]

Question continues on next page.

(c) Consider now a filter bank specified by the following signal equations:

$$\begin{aligned} y_0 &= D_3G_0D_3G_0x \\ y_1 &= D_3G_1D_3G_0x \\ y_2 &= D_3G_2D_3G_0x \\ y_3 &= D_3G_1x \\ y_4 &= D_3G_2x. \end{aligned}$$

i. Draw the equivalent single-level five-channel filter bank [4]

ii. Assume that $G_0(z) = (1 + z + z^2)/\sqrt{3}$, $G_1(z) = (-1 + 2z - z^2)/\sqrt{6}$ and that $G_2(z) = (1-z^2)/\sqrt{2}$, also assume that $x[n] = 1$ and ignore any boundary effect. Which of the signals $y_0[n]$,$y_1[n]$, $y_2[n]$, $y_3[n]$ and $y_4[n]$ is nonzero? (Justify your answer). [3]

iii. Assume now that $x[n] = n$ and again ignore any boundary effect. Which of the signals $y_0[n]$,$y_1[n]$, $y_2[n]$, $y_3[n]$ and $y_4[n]$ is nonzero? (Justify your answer). [3]

3. (a) Consider the scaling function

$$\varphi(t) = \frac{\sin(\pi t)}{\pi t}.$$

i. Show that the system

$$\varphi(t+k) = \frac{\sin(\pi(t+k))}{\pi(t+k)} \qquad k \in \mathbb{Z}$$

forms an orthonormal basis. [5]

ii. Show that $\varphi(t)$ satisfies partition of unity:

$$\sum_{n=-\infty}^{\infty} \varphi(t-n) = 1$$

(Hint: Use Poisson summation formula). [5]

iii. One can show that the function

$$\varphi(t) = \frac{\sin(\pi t)}{\pi t}$$

satisfies a two-scale equation:

$$\varphi(t) = \sqrt{2} \sum_n g_0[n] \varphi(2t-n).$$

A. By using the orthogonality condition $\langle \varphi(t), \varphi(t-k) \rangle = \delta_k$, show that

$$g_0[n] = \sqrt{2} \langle \varphi(t), \varphi(2t-n) \rangle.$$

[5]

B. Find the exact expression for $g_0[n]$ when $\varphi(t) = \frac{\sin(\pi t)}{\pi t}$.
(Hint: Use Parseval identity: $\langle f(t), g(t) \rangle = \frac{1}{2\pi} \langle \hat{f}(\omega), \hat{g}(\omega) \rangle$). [5]

iv. Write the exact expression for

$$\psi(t) = \sqrt{2} \sum_n g_1[n] \varphi(2t-n).$$

[5]

4. Suppose you are given a two-channel FIR filter bank with real coefficients and synthesis lowpass filter

$$g_0[n] = \frac{1}{2\sqrt{2}}(\delta_{n+1} + 2\delta_n + \delta_{n-1}).$$

(a) Is it possible that you were given an orthogonal filter bank? Briefly justify your answer. [5]

(b) Find a synthesis highpass filter $G_1[n]$ with linear phase and length greater than 1 such that, with the appropriate analysis filters, a perfect reconstruction filter bank is formed. (The solution is not unique.) [8]

(c) Consider the equivalent filter

$$G_0^{(i)}(z) = \prod_{k=0}^{i-1} G_0(z^{2^k})$$

obtained by iterating the filter bank decomposition i times. Consider the function

$$\varphi^{(i)}(t) = 2^{i/2} g_0^{(i)}[n], \qquad n/2^i \le t < (n+1)/2^i.$$

i. Can you say anything about the convergence of $\lim_{i\to\infty} \varphi^{(i)}(t)$? [4]

ii. Assume that $\lim_{i\to\infty} \varphi^{(i)}(t)$ exists and assume that the limit converges to the function $\varphi(t)$. We know that, in the case of convergence, $\varphi(t)$ is a valid scaling function. Can you say anything about continuity of $\varphi(t)$? [4]

iii. State the number of vanishing moments of the analysis wavelet function obtained from $\varphi(t)$. [4]